SEC P vs NP log 2026-05-12

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-12 04:30 UTC

SEC P vs NP — daily log 2026-05-12

9 cycles recorded

Free Cumulant Sum Bounded by Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability Theory × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-12 01:18 UTC
- **Entry ID**: `ecd8f062c47d`

Statement

For a |X|×|Y| matrix M defining a partial function f, let μ(M) = ∑_{k≥1} |c_k(M)| where c_k(M) are the free cumulants of M's entries. Then μ(M) ≤ O(√(R(f))) where R(f) is the randomized communication complexity of f. For DISJ_n, μ(DISJ_n) = Ω(n) holds.

Rationale

Free cumulants capture non-commutative independence structures in matrix entries, which may correlate with the information-theoretic cost of coordinating disjointness checks. The sum's growth rate could reflect the minimal communication required to resolve entangled dependencies.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38ecf086.py", line 75, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38ecf086.py", line 54, in run_trial
    cumulants = free_cumulants(M)
                ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38ecf086.py", line 39, in free_cumulant
   R[i][j] -= sum(R[x][y] * R[i - x][j - y] for x in range(i) for y in range(j)) / (i * j)
            ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^~~~~~~~~
ZeroDivisionError: division by zero
```

Judge reasoning

Test crashed due to division by zero, preventing evaluation of conjecture | next: Fix implementation to handle zero division and retest with validated inputs

---

Plethysm Coefficient Gap in Permanent vs Determinant Decompositions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl Duality × Boolean Circuit Size for Permanent Polynomial
- **Recorded**: 2026-05-12 01:34 UTC
- **Entry ID**: `40591186a3ea`

Statement

For all $n \geq 2$, the plethysm coefficient λ(perm_n) of the symmetric power S^k(perm_n) exceeds λ(det_m) + $\varepsilon \cdot n^2$ for m < n^{1.5}, where $\varepsilon > 0$ is universal. This gap persists under linear substitutions preserving the polynomial's symmetry type.

Rationale

Plethysm coefficients encode symmetry constraints of polynomial representations. By linking them to circuit complexity, we may expose algebraic barriers to expressing permanent as a determinant-like circuit, leveraging representation-theoretic invariants resistant to natural proofs.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.19s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d9d81969.py", line 81, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d9d81969.py", line 58, in run_trial
    if lambda_perm_n > lambda_det_m + epsilon * n ** 2:
                       ~~~~~~~~~~~~~^~~~~~~~~~~~~~~~~~
OverflowError: int too large to convert to float
```

Judge reasoning

Test crashed with overflow error before producing data; cannot verify conjecture's validity | next: Implement arbitrary-precision arithmetic or numerical stability checks for large n

---

Symmetric Group Fourier Coefficients Inverse Proportional to ACC^0 Circuit Size for Sipser Function

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Finite Groups × ACC^0 Circuit Size for Sipser Function
- **Recorded**: 2026-05-12 01:48 UTC
- **Entry ID**: `567baddf9ba5`

Statement

For the Sipser function S_n on n variables, the maximum absolute Fourier coefficient over the symmetric group S_n's irreducible representations is Θ(1/k), where k is the minimal ACC^0 circuit size computing S_n. This holds for all n ≥ 2.

Rationale

Symmetric group representations capture combinatorial symmetries of permutation-based functions like Sipser's. Fourier coefficients over these representations quantify 'non-abelian harmonic analysis' of the function, potentially revealing structural obstructions to efficient ACC^0 computation.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_136fa469.py", line 80, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_136fa469.py", line 59, in run_trial
    for k in range(1, 2**n + 1):
                         ^
NameError: name 'n' is not defined
```

Judge reasoning

Test crashed due to undefined variable 'n' before producing results | next: Fix the NameError in test code by defining 'n' before usage

---

Integer Point Count in Solution Polytope Bounds Extended Frege Proof Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometry of Numbers × Extended Frege Proof Length
- **Recorded**: 2026-05-12 01:54 UTC
- **Entry ID**: `4798bdb0751b`

Statement

For a 3-SAT instance Φ with n variables, the number of integer solutions S(Φ) satisfies S(Φ) ≥ 2^{n - L(Φ)}, where L(Φ) is the length of the Extended Frege proof of Φ. Equality holds when Φ is a tautology with a unique solution.

Rationale

The geometry of numbers connects the combinatorial structure of integer solutions to algebraic constraints, while Extended Frege proofs formalize logical deductions. The conjecture posits that solution density directly impacts proof complexity via exponential trade-offs, exposing hidden geometric regularities in tautology refutations.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8072c074.py", line 82, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8072c074.py", line 64, in run_trial
    S_phi = sum(1 for _ in filter(dpll, [sat_formula]))
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8072c074.py", line 64, in <genexpr>
    S_phi = sum(1 for _ in filter(dpll, [sat_formula]))
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8072c074.py", line 27, in dpll
    literal, rest = sat_formula[0]
    ^^^^^^^^^^^^^
ValueError: too many values to unpack (expected 2)
```

Judge reasoning

Test crashed during execution, preventing data collection. The error suggests a structural issue with the sat_formula handling in the DPLL algorithm. | next: Debug the DPLL implementation's handling of sat_formula structure to resolve the unpacking error

---

Matroid Rank Bounded by Nisan-Wigderson Seed Length for 3-SAT

- **Verdict**: INCONCLUSIVE
- **Bridge**: Matroid Theory × Nisan-Wigderson PRG Seed Length
- **Recorded**: 2026-05-12 02:14 UTC
- **Entry ID**: fa70962e79f5

Statement

For any 3-SAT instance with n variables, the seed length of the Nisan-Wigderson PRG fooling it is at most the rank of the matroid formed by its clause hypergraph, where the rank is the maximal size of a set of clauses with pairwise disjoint variable support.

Rationale

Matroid rank captures combinatorial dependencies between clauses, which may constrain the pseudorandomness needed to fool the formula. This bridges structural properties of CNF formulas with PRG efficiency, leveraging matroid theory's role in encoding combinatorial constraints.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.01s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Run test with extended timeout and memory limits to complete execution

---

Matroid Rank Submodularity and Monotone Circuit Size for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory × Monotone Circuit Size for k-CLIQUE
- **Recorded**: 2026-05-12 02:20 UTC
- **Entry ID**: `b052cc185466`

Statement

For a k-CLIQUE CNF formula Φ on n variables, the rank function r(S) of the matroid M constructed from Φ satisfies r(S) ≥ Ω(n^{1/2}) for any subset S of size Ω(n^{1/2}), and the minimum number of elements needed to achieve r(S) = Ω(n^{1/2}) is Ω(n^{1/2}).

Rationale

The submodularity of the rank function in matroid theory could mirror the structure of monotone circuits, where dependencies between clauses (elements) affect the circuit's size. By linking the rank's growth rate to the circuit's complexity, we might uncover structural constraints.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Test crashed before producing data, preventing evaluation of support fraction or counterexamples. | next: Re-run test with detailed logging to identify crash cause and collect empirical evidence

---

Symplectic Rank of Disjointness Communication Matrix Bounds Randomized Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Symplectic Geometry × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-12 02:46 UTC
- **Entry ID**: `8acf355ada09`

Statement

The symplectic rank of the DISJOINTNESS communication matrix M is Ω(n), where the symplectic rank is defined as the minimal k such that M decomposes into k rank-1 symplectic tensors under a fixed symplectic form. This lower bounds the randomized communication complexity of DISJOINTNESS.

Rationale

Symplectic geometry provides invariants that capture structural properties of matrices, which may correlate with communication complexity. The symplectic rank, as a decomposition invariant, could expose hidden algebraic constraints in the matrix that mirror information-theoretic barriers.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dfb495a0.py", line 102, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dfb495a0.py", line 74, in run_trial
    if not is_symplectic_form(M, n):
           ^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dfb495a0.py", line 44, in is_symplectic
   return matrix_multiply(matrix_multiply(M, M), I) == -I and matrix_multiply(transpose(M), M) == -
I
                                                         ^^
TypeError: bad operand type for unary -: 'list'
```

Judge reasoning

Test crashed due to type error in matrix operations, preventing evaluation of conjecture | next: Fix matrix arithmetic implementation to handle symplectic form verification correctly

---

Schur Coefficient Density Inverse Proportional to Frege Proof Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Symmetric Function Theory × Frege Proof Length
- **Recorded**: 2026-05-12 03:10 UTC
- **Entry ID**: `2b3f03ad1b52`

Statement

For a homogeneous polynomial f of degree d in n variables, let S(f) denote the number of non-zero coefficients in its Schur decomposition. Then, the Frege proof length of f = 0 is at least Ω(S(f)^{1/2})

Rationale

The density of Schur coefficients reflects the complexity of the polynomial's representation, which might correlate with the difficulty of proving its identity through Frege systems. Symmetric function theory provides a framework to decompose polynomials into irreducible components, exposing structural properties that could influence proof complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
ue': 1, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': ''}
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'Frege Proof Length', 'metric_value': 1, 'instances_tested': 1, 'conjecture_ho

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_98998d42.py", line 121, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_98998d42.py", line 121, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                              ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Test crashed due to KeyError before producing reliable data | next: Fix the test code to handle missing 'seed' field and re-run with proper error handling

---

Free Cumulant Sum Bounded by Log-Size for Read-Twice BPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability Theory × Read-Twice Branching Programs
- **Recorded**: 2026-05-12 03:25 UTC
- **Entry ID**: `93b47d86db79`

Statement

For any read-twice BP P computing IP_2, the sum of free cumulants Σ_{k=1}^∞ κ_k(P) ≤ log(size(P)) + O(1). For random read-twice BPs over GF(2), Σ κ_k(P) = Θ(log n).

Rationale

Free cumulants capture non-commutative independence structures; their sum could distinguish read-twice BPs (with limited independence) from unrestricted BPs (with full independence). IP_2's inherent symmetry might force linear cumulant growth.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.01s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out without producing supporting data or counterexamples | next: Increase timeout duration and re-run tests with extended computation limits